

30 Hydrocarbons

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Aromatic reactions commonly proceed by electrophilic substitution to preserve benzene's delocalised π system. Nitration uses nitronium ion generated by concentrated nitric and sulfuric acids; Friedel-Crafts reactions use a catalyst. Electron-donating/withdrawing substituents alter ring reactivity and direct substitution position.

Worked example: follow the reasoning

Benzene reacts with NO_2^+ to form nitrobenzene via a sigma complex followed by loss of H^+ , restoring aromaticity.

Quick check · multiple choice

A What is the electrophile in benzene nitration?

[1]

- A NO_3^-
- B NO_2^+
- C NH_4^+
- D OH^-

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Name electrophile in benzene nitration and explain why substitution predominates over addition.

[3]

Answers and explanation

Quick check: B. NO_2^+

Question 1 [3 marks]

NO_2^+ [1]; addition would destroy the stable delocalised aromatic system [1], while substitution restores it after deprotonation [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.